Retail Order Data Analysis

Mini Project Report

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Batch: MDTM38[DS-S-WD-T-B33]

Tools Used

- Kaggle API
- Python (Pandas)
- MySQL
- Streamlit
- Plotly

Problem Statement

- This project analyzes retail sales data to identify patterns in revenue, profitability, regional performance, and product sales using SQL and Python, and visualizes insights through an interactive Streamlit dashboard.

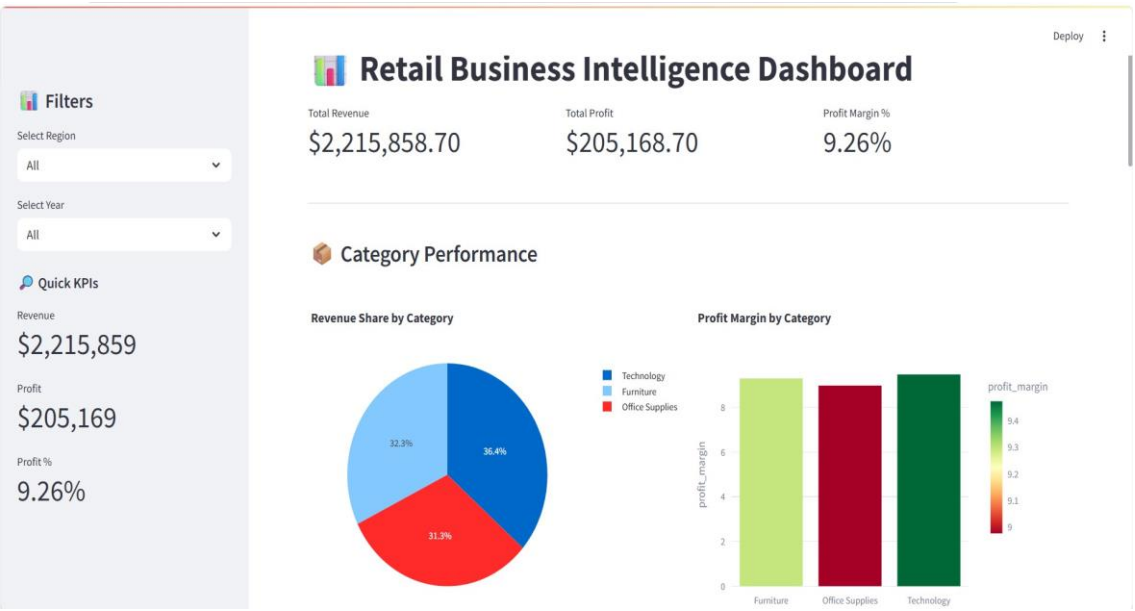
Data Cleaning Steps

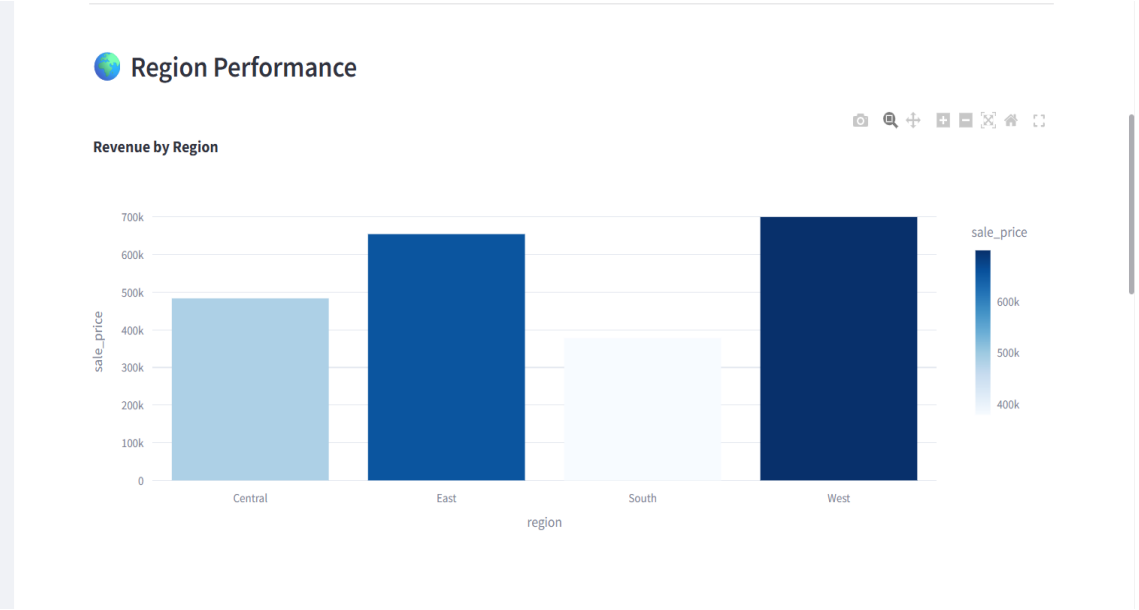
- Standardized column names.
- Converted discount_percent values from percentage to decimal.
- Created sale_price column using list_price and discount_percent.
- Calculated profit column as sale_price minus cost_price.
- Converted order_date column to datetime format.
- Validated numeric fields and removed inconsistencies.

SQL Queries and Business Insights

Query	Insight
Total Revenue	2215858.7 total revenue generated.
Total Profit	205168.7 total profit generated.
Total Orders	9994 orders are processed.
Revenue by Category	Technology contributes to the highest revenue. Profit by
Category	Technology category generates the most profit. Average
Discount by Category	Discount distribution differs across categories. Revenue by
Region	West region produces the highest revenue.
Profit by Region	West region produces the highest profit. Average
Discount by Region	Regional discount patterns is similar.
Top 10 Products by Revenue	Few products dominate revenue share.
Top 5 Products by Profit	High Profit items should be prioritized.
High Discount Products	Large discounts reduce profit margin.
Revenue by Year	Revenue increased from 2022 to 2023.
Profit by Year	Profit trend shows growth.
Revenue by Month	Monthly revenue shows seasonal variation.
Profit Margin by Category	Technology has the strongest margin.
Segment Quantity Sold	Consumer segment drives the largest volume.
Most Profitable Month	Certain months outperform others.
Highest Avg Sale Price Region	Some regions maintain higher pricing.
Product Revenue Ranking	Ranking highlights top products.

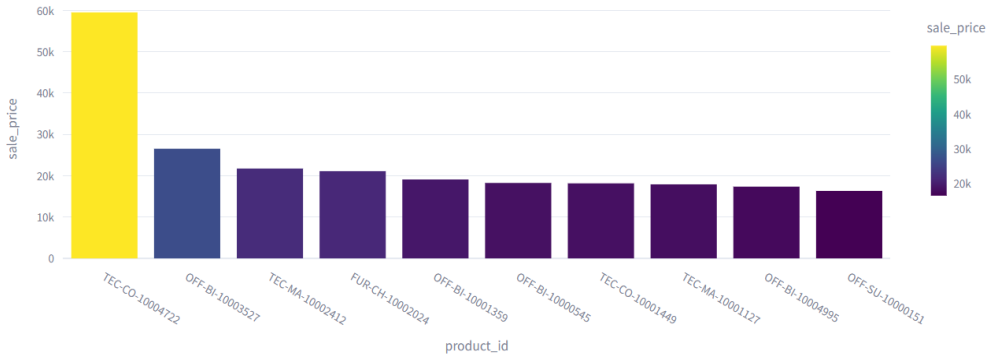
Dashboard Screenshots





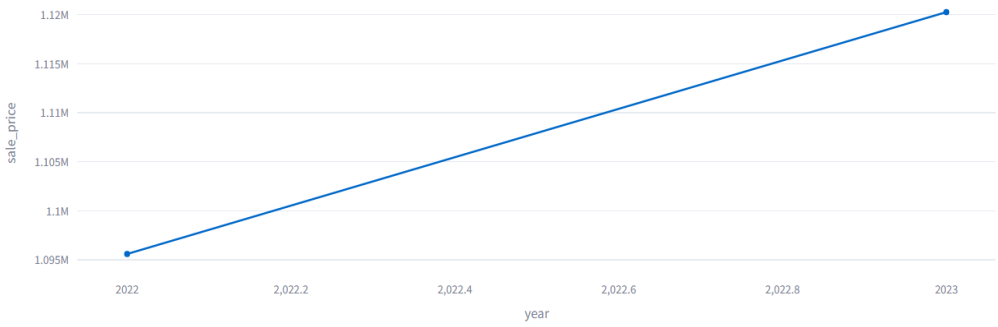
🏆 Top 10 Products by Revenue

Top 10 Revenue Products



📅 Revenue Trend

Year-wise Revenue Trend



Conclusion

- The analysis shows that Technology products and the West region generate the highest revenue and profit. Revenue trends indicate consistent growth from 2022 to 2023. The Streamlit dashboard enables interactive exploration of these insights and supports data driven decision making.